



TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

World's first 'true' dynamic holographic display chips

Swave Photonics has developed the world's first 3D holographic display technology tailored for compact extended reality (XR) devices



like smart glasses. Their Holographic eXtended Reality (HXR) technology features the world's smallest

pixel—under 300nm—allowing it to create high-resolution, realistic images by manipulating light. This chip-based technology integrates AI to blend digital and physical realities, enhancing user experience. Utilising phase change material (PCM) on a standard CMOS process, it achieves genuine 3D holography with up to 64 gigapixel resolution. Aimed at producing affordable, lightweight AR smart glasses with all-day battery life, the chip supports compact, wearable form factors and works seamlessly with AI services like image recognition. It is designed for power efficiency and cost-effectiveness, using established semiconductor processes for quick commercialisation.

Swave Photonics
<https://swave.io>

Innovative counter-rotating motors

Vector Technics has introduced the 'eVector' counter-rotating motors, designed to significantly enhance the performance and efficiency of unmanned aerial vehicles (UAVs). Utilising counter-rotating technology, these motors are setting new industry standards and are expected

to be adopted across various sectors. Industries that stand to benefit include commercial sectors such as aerial photography and surveying; defence and security for improved surveillance and reconnaissance; research and

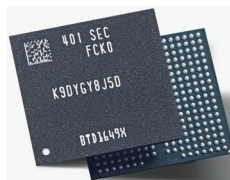


development organisations focused on UAV technological advancements; and aerospace companies seeking to incorporate new propulsion technologies to expand UAV functionality.

Vector Technics
<https://vectortechnics.com>

Industry's first mass production of 9th-gen V-NAND

Samsung Electronics has announced the start of mass production for its one-terabit (Tb) triple-level cell (TLC) 9th-generation vertical NAND (V-NAND). The V-NAND features the industry's smallest cell size and thinnest mould, achieving a 50% increase in bit density over the 8th generation. Innovations



such as cell interference avoidance and life extension techniques have improved the product's quality and reliability, while the elimination of dummy channel holes has reduced the memory cells' planar area. It incorporates the new 'Toggle 5.1' NAND flash interface, boosting data input/output speeds by 33% to 3.2 gigabits-per-second (Gbps) and

supporting expansions like PCIe 5.0. Additionally, a 10% improvement in power consumption over the previous generation, aligning with the growing importance of reducing energy usage and carbon emissions, makes this V-NAND an optimal choice for future applications.

Samsung Electronics
<https://news.samsung.com>

Ultra-efficient wireless SoC family

Silicon Labs has launched wireless system-on-chip (SoC) components designed for devices running on harvested energy, eliminating the need for batteries. The xG22 targets the



'ambient IoT' market, where devices draw power from radio waves, light, heat, vibrations, and other movements. These

SoCs can wake from a zero-energy cold start to transmit in eight milliseconds using 150µJ of energy. From deep sleep, they wake using 16.6µJ in 1.83 milliseconds, 78 percent less energy than similar microcontrollers. Suitable for IoT manufacturers, smart home firms, wearable tech developers, and environmental monitoring companies, these SoCs also benefit healthcare, industrial automation, asset tracking, and consumer electronics by enabling battery-free devices. Research labs can use them for energy harvesting and sustainable technology projects.

Silicon Labs
<https://www.silabs.com>

World's first quad-port SSD

Micron Technology has introduced the automotive-grade Micron 4150AT